Creating and using Variables

Key Stage 3: Year 7

Cian Lyons
04/03/2025

Understanding variables

Sometimes we need computers to remember the *information* we give it and that it calculates during programs.

A variable can be thought of as a box that the computer can use to store a value.

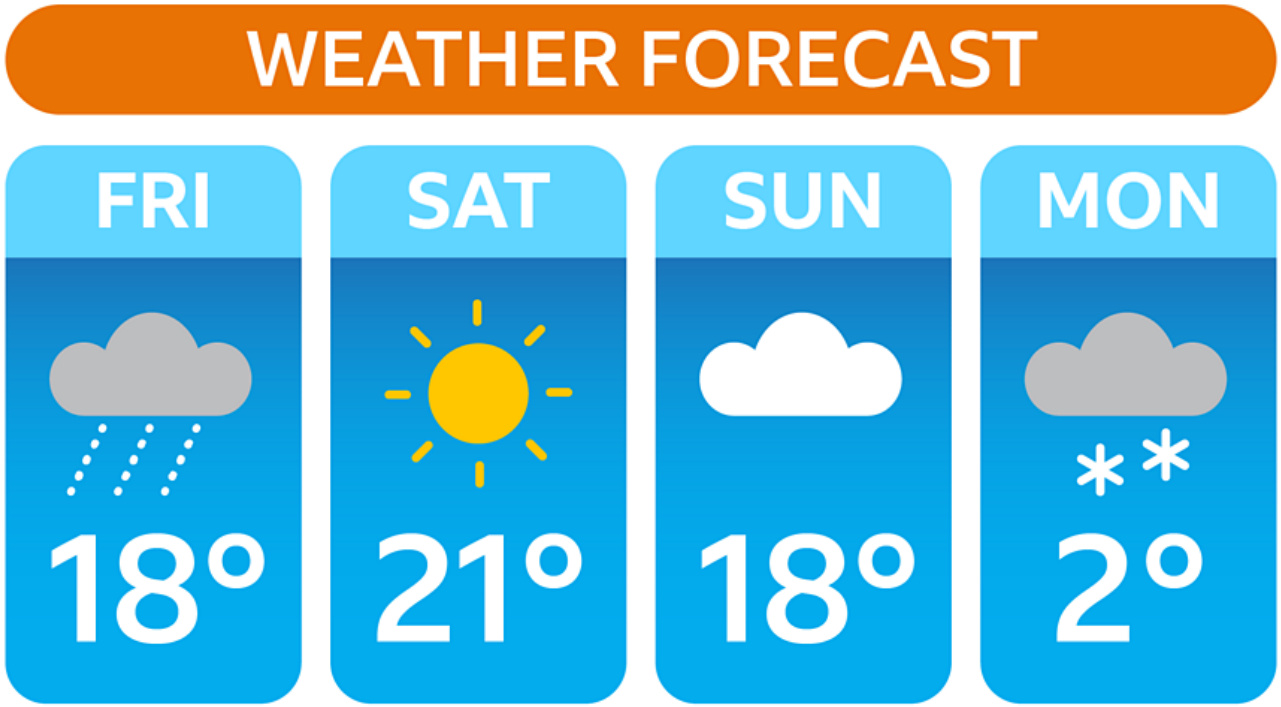
The value held in that box can change or ‘vary’. A program can use as many variables as it needs it to.

All variables are made up of three parts:

- a name
- a type
- a value

Why use variables? Do many things vary?

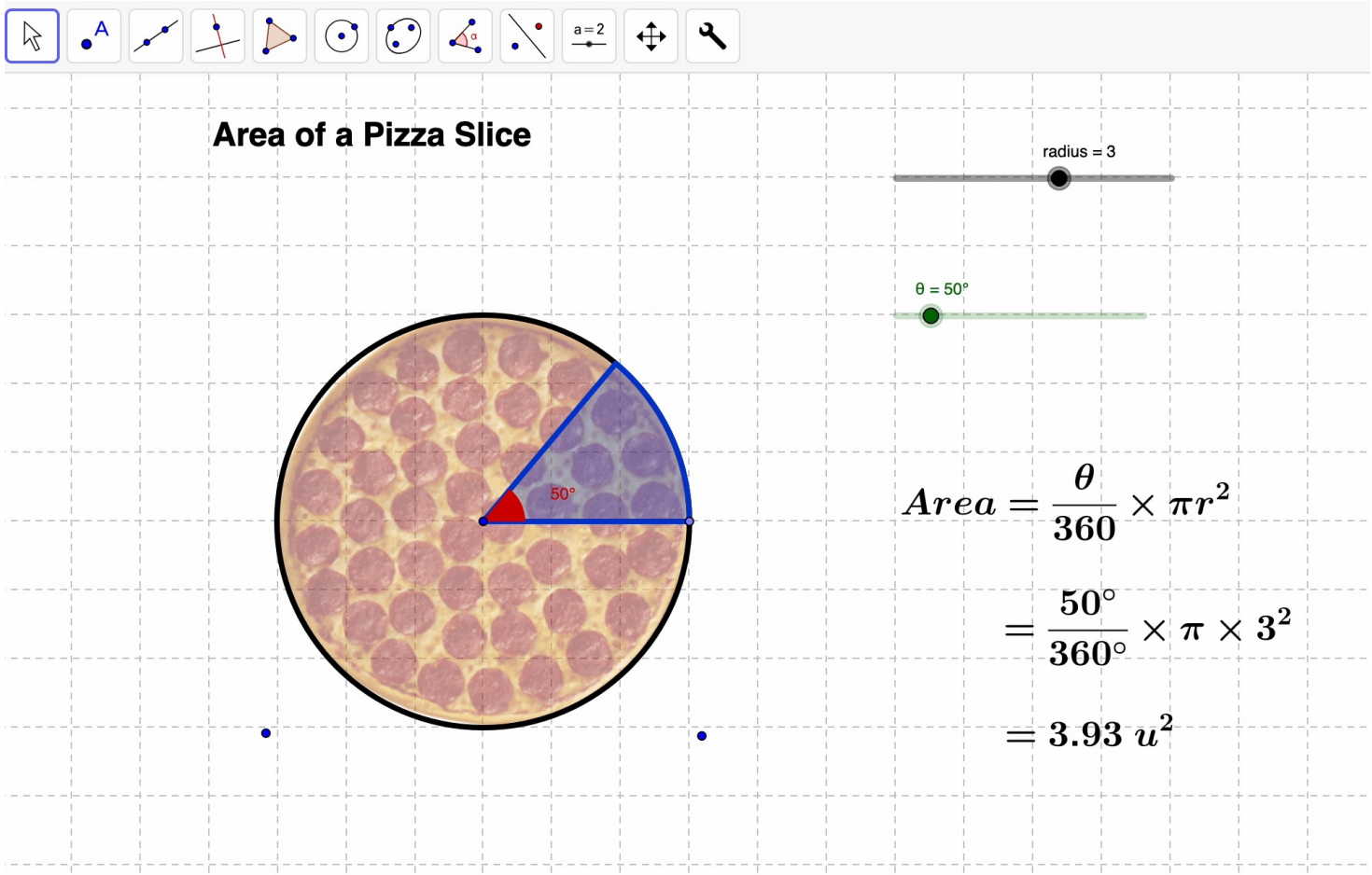
Variables in life



Variables in life

Variables in life

Variables in life

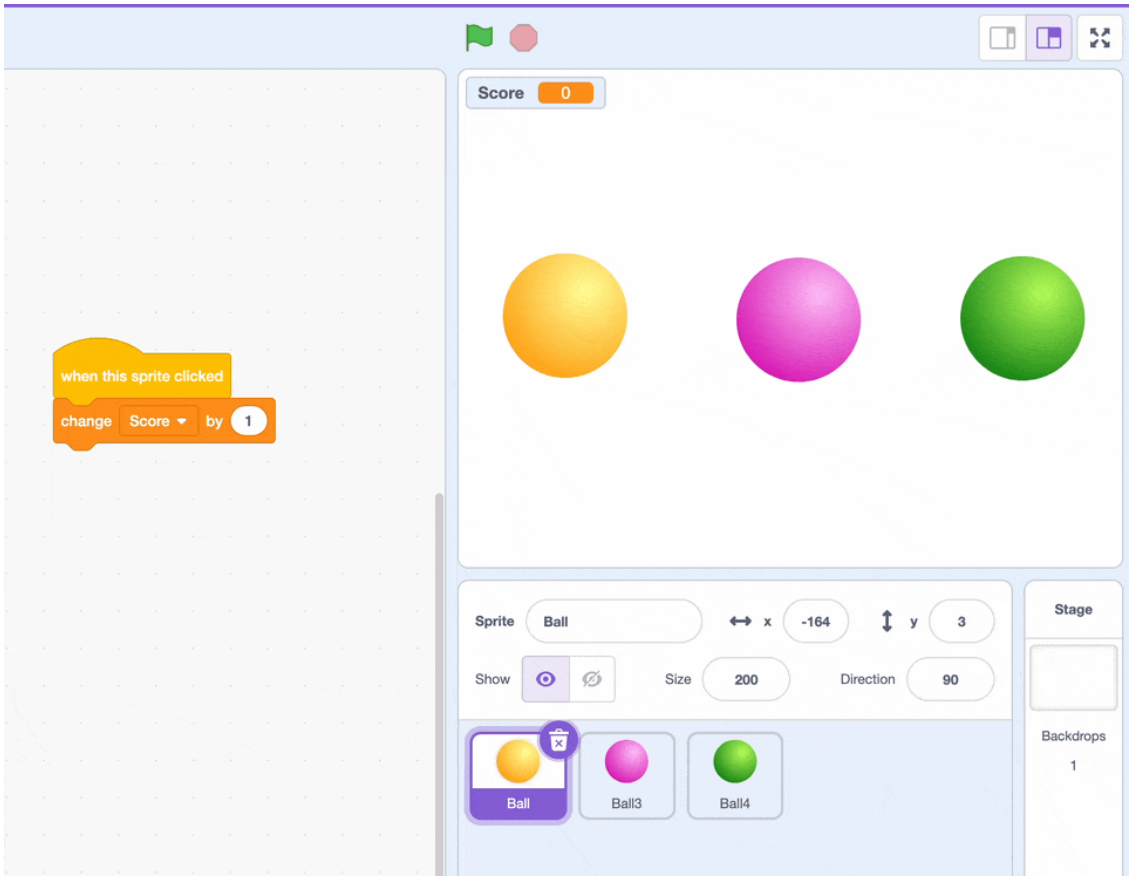


Variables in Scratch

What changes in this project?

When does it change?

Is the change always the same?



Variable types in programming

String	Float or Real	Integer	Boolean
Title	Rating	Times Viewed	Favourite
Zombie Attack	9.5	83	True
True Love	8.0	5	True
Mission: Pluto	2.5	1	False

Typed vs Untyped Programming Languages

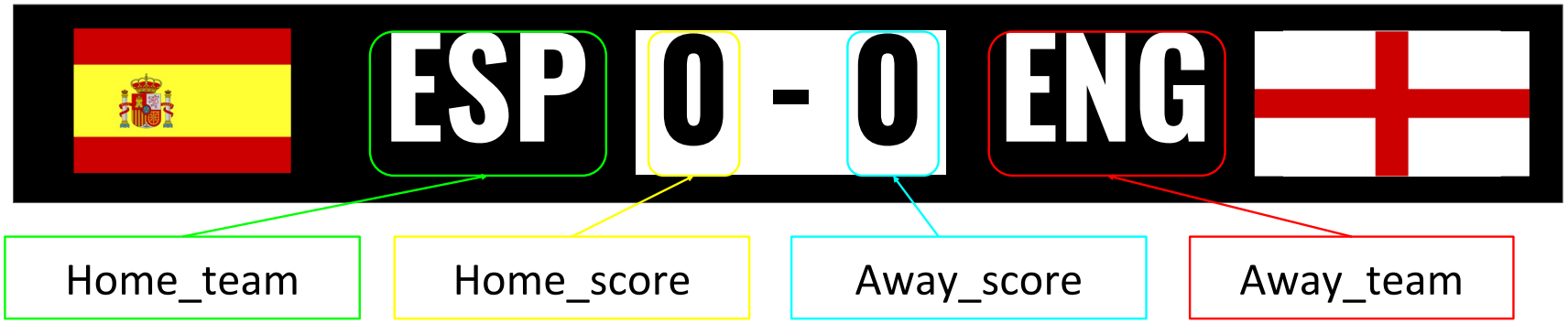
```python
# Untyped language: Python
name = "Cian"
age = 30

print(f"Hi, my name is {name}, I'm {age}")
```

```java
// Typed language: Java
String name = "Cian"
int age = 30

String output = "Hi, my name is " + name + ", I'm " + age

System.out.print(output)
```

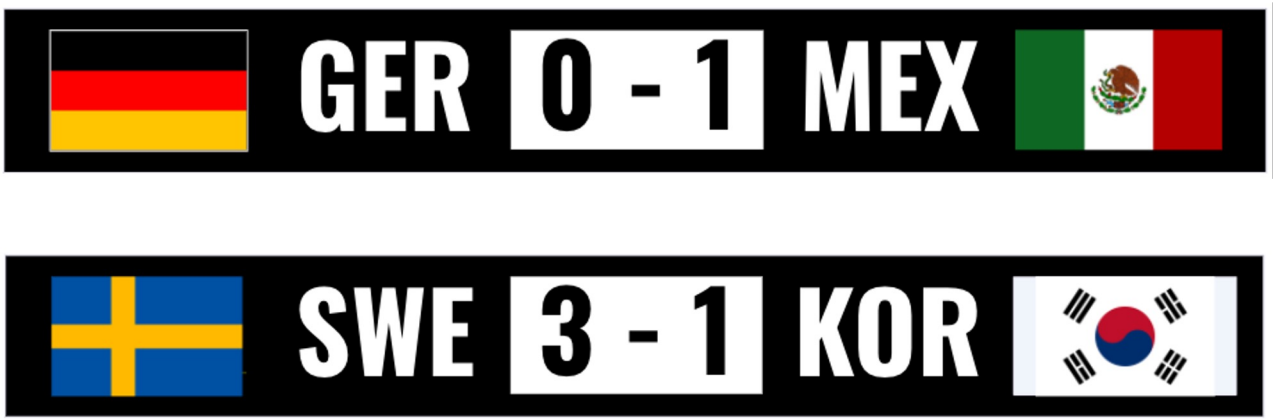
Worksheet Q1

In the Scratch example we looked at earlier you identified the variable Score.

More complicated situations might involve more than one variable:



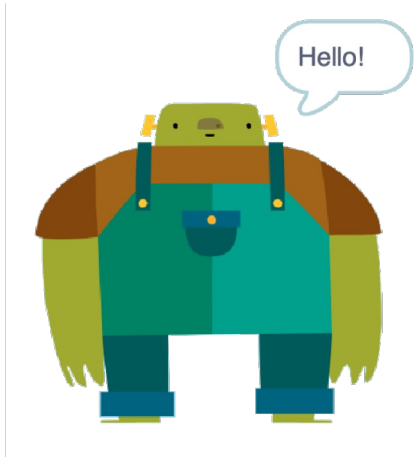
On the activity sheet, find the values of the variables.



Worksheet Q2

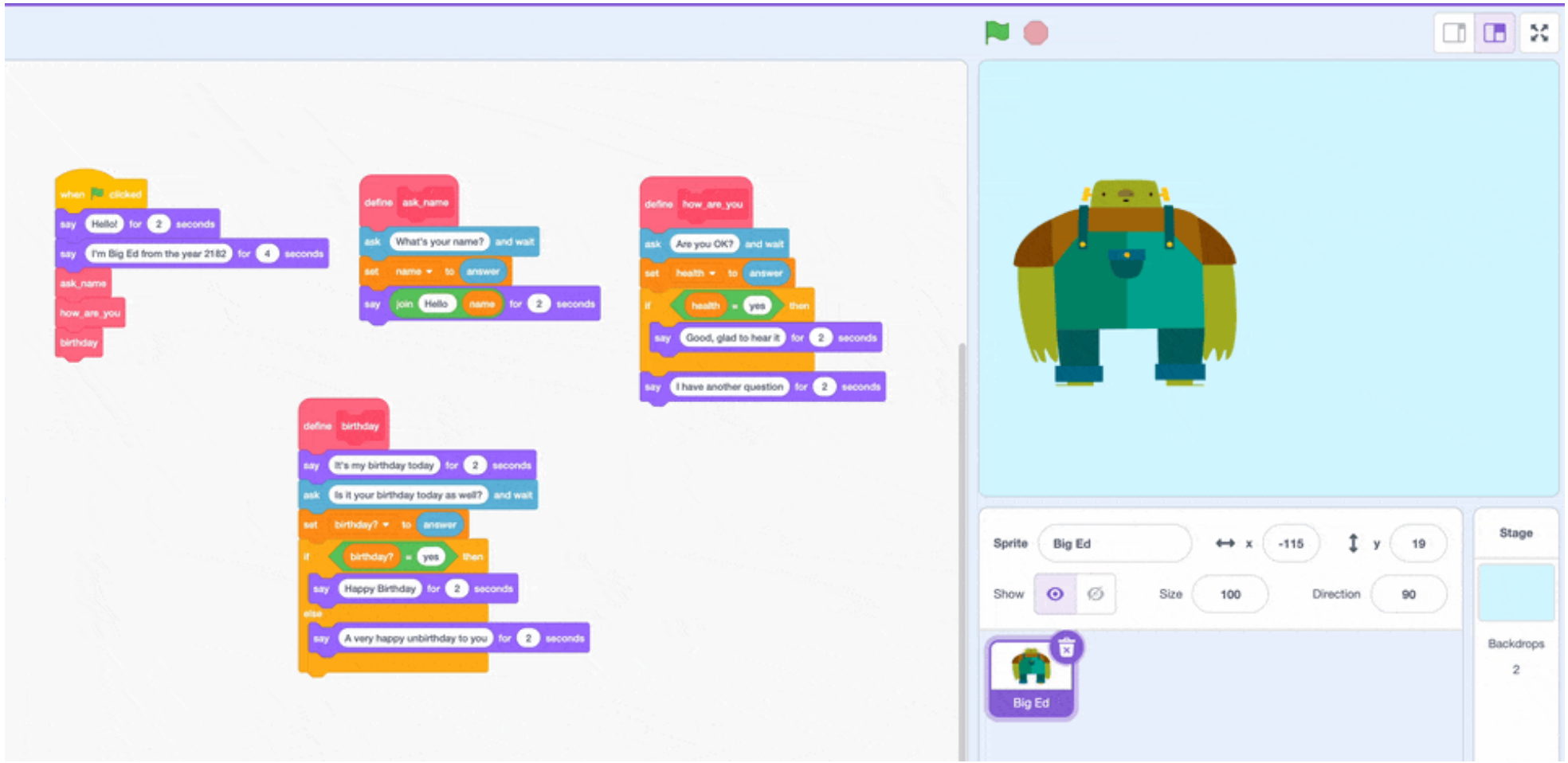
Big Ed is your friendly chatbot.

This code has been added to Big Ed in Scratch:

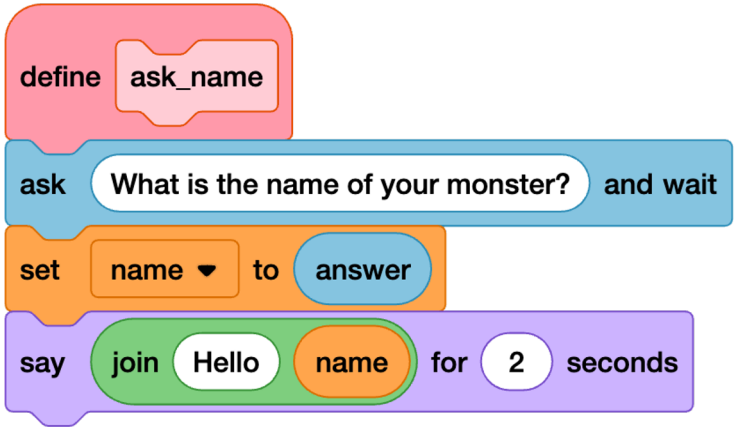



```
when [flag] clicked
say [hello] for [2] seconds
say [I'm Big Ed from the year 2182] for [2] seconds
ask_name
```

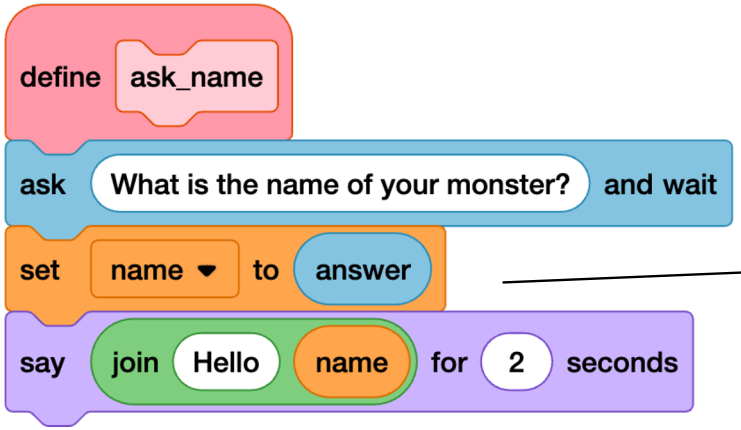
```
define ask_name
ask [What is the name of your monster?] and wait
set [name ▾] to [answer]
say [join [Hello] [name]] for [2] seconds
```


when clicked
say Hello! for 2 seconds
say I'm Big Ed from the year 2182 for 4 seconds
ask_name
how_are_you
birthday
define ask_name
ask What's your name? and wait
set name to answer
say join Hello name for 2 seconds
define how_are_you
ask Are you OK? and wait
set health to answer
if health = yes then
say Good, glad to hear it for 2 seconds
say I have another question for 2 seconds
define birthday
say It's my birthday today for 2 seconds
ask Is it your birthday today as well? and wait
set birthday? to answer
if birthday? = yes then
say Happy Birthday for 2 seconds
else
say A very happy unbirthday to you for 2 seconds
Sprite Big Ed
x -115
y 19
Show
Size 100
Direction 90
Big Ed
Stage
Backdrops
2

What are the two **variables** in this code block?



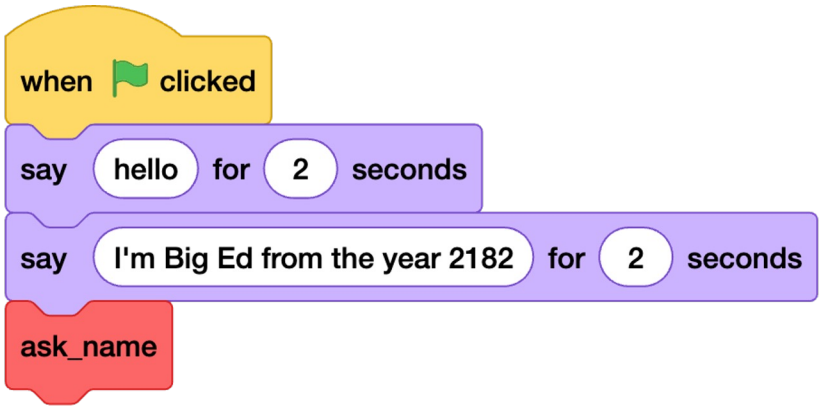
What are the two **variables** in this code block?

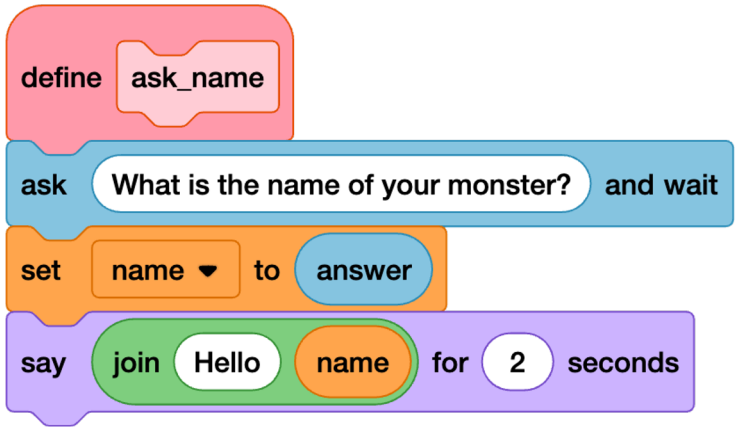


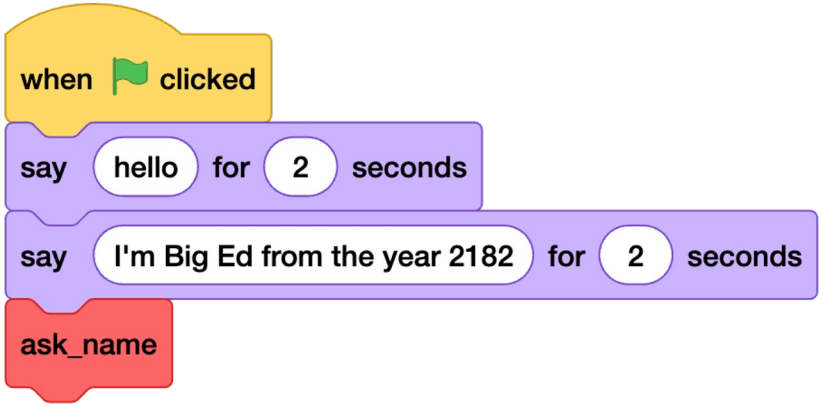
name and answer

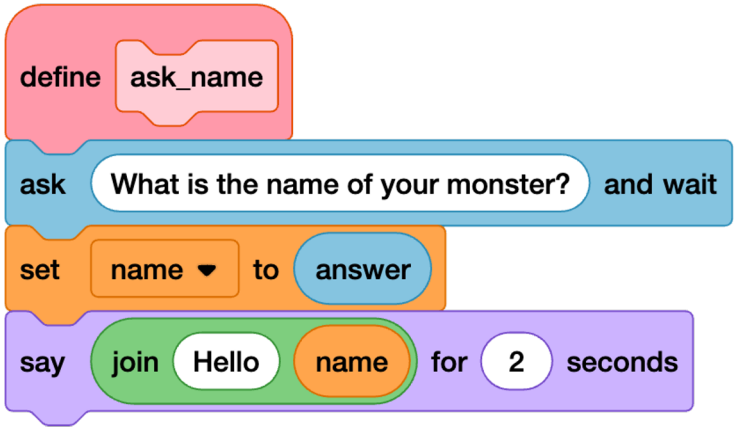
‘answer’ is a built-in **variable** in Scratch that holds the **value** a user types in.
The **variable** ‘name’ has been created by the programmer.

Predict what will happen when this code is run.









The Big Ed sprite will say hello and introduce themselves. It will then call the ask_name function / subroutine.

The ask_name subroutine will run a sequence of commands that asks the user for the name of their monster, stores that name in a **variable**, before outputting it in a hello {name} message.

How confident are you? (1–3)

- I can identify examples of information that is variable
- I can explain that the way that a variable changes can be defined
- I can identify that variables can hold numbers or letters

3 – Very confident

2 – Unsure

1 – Not confident

As **variables** can change during a program, it is useful for a programmer to **trace** the **value** of the **variable**.

This means the programmer will know exactly what the **value** of the **variable** will be at each stage.

Unexpected program behaviour or output might require us to **debug**, tracing is a useful skill to debug.

Worksheet Q3

Big Ed has just arrived on a new planet and he's measuring the temperature of his new environment.

Is the temperature likely to change?

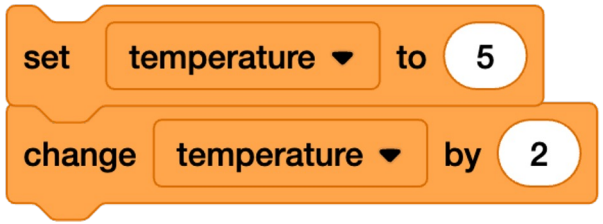
Yes, the temperature will change, so the **variable** that holds the temperature will need to be updated.

Variables can be updated in Scratch by using two different blocks.

You can update the **value** a **variable** holds by using the set block.

You can also use the change by block. This block will change the temperature **value** that is stored.

Trace the **value** of temperature, what **value** does temperature hold at the end of this code block?



a 5

b 3

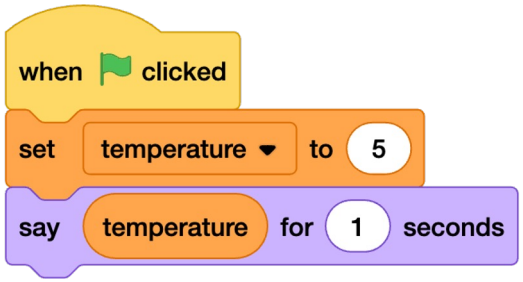
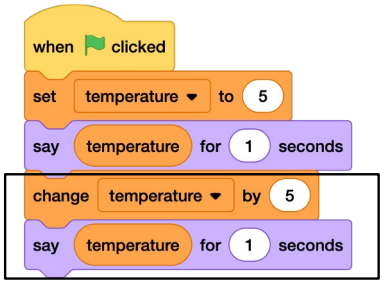
c 7 ✓

temperature starts at 5 and is then changed by 2, so becomes 7.

The following code was added to the Big Ed Scratch program:

```
when [flag] clicked
set temperature to 5
say temperature for 1 seconds
change temperature by 5
say temperature for 1 seconds
change temperature by -2
say temperature for 1 seconds
set temperature to 0
say temperature for 1 seconds
change temperature by 6
say temperature for 1 seconds
```

Trace the **value** of temperature at each stage.

stage	code	**value** of temperature
a)		
b)		

Trace the **value** of temperature at each stage.

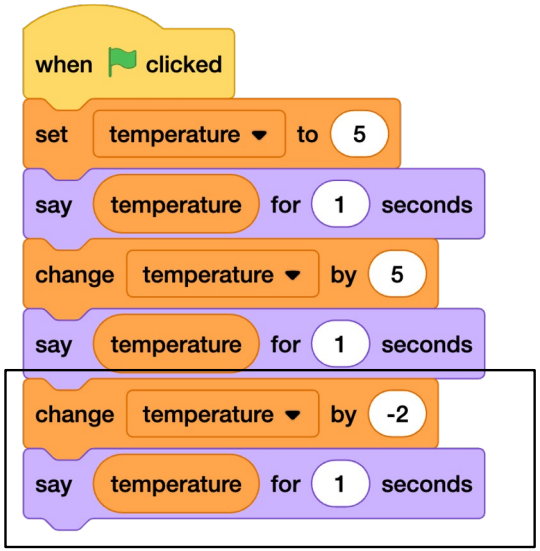
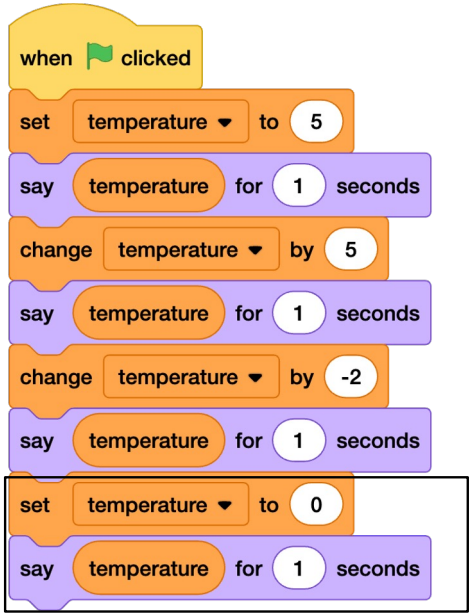
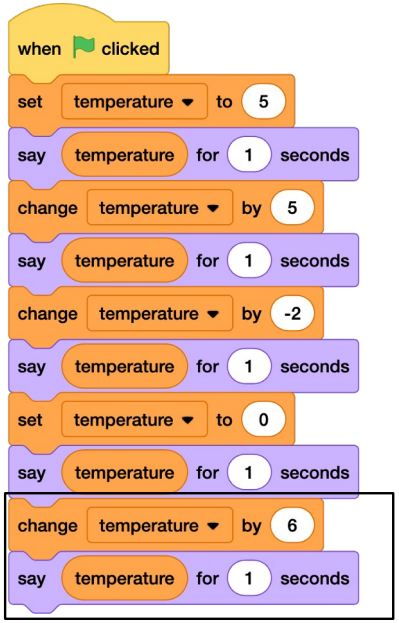
stage	code	**value** of temperature
c)		

Image: Scratch Foundation under CC BY-SA 2.0

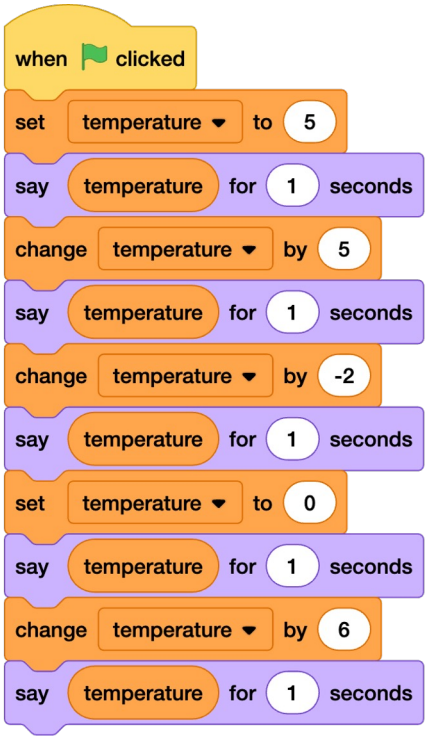
Trace the **value** of temperature at each stage.

stage	code	**value** of temperature
d)		

Trace the **value** of temperature at each stage.

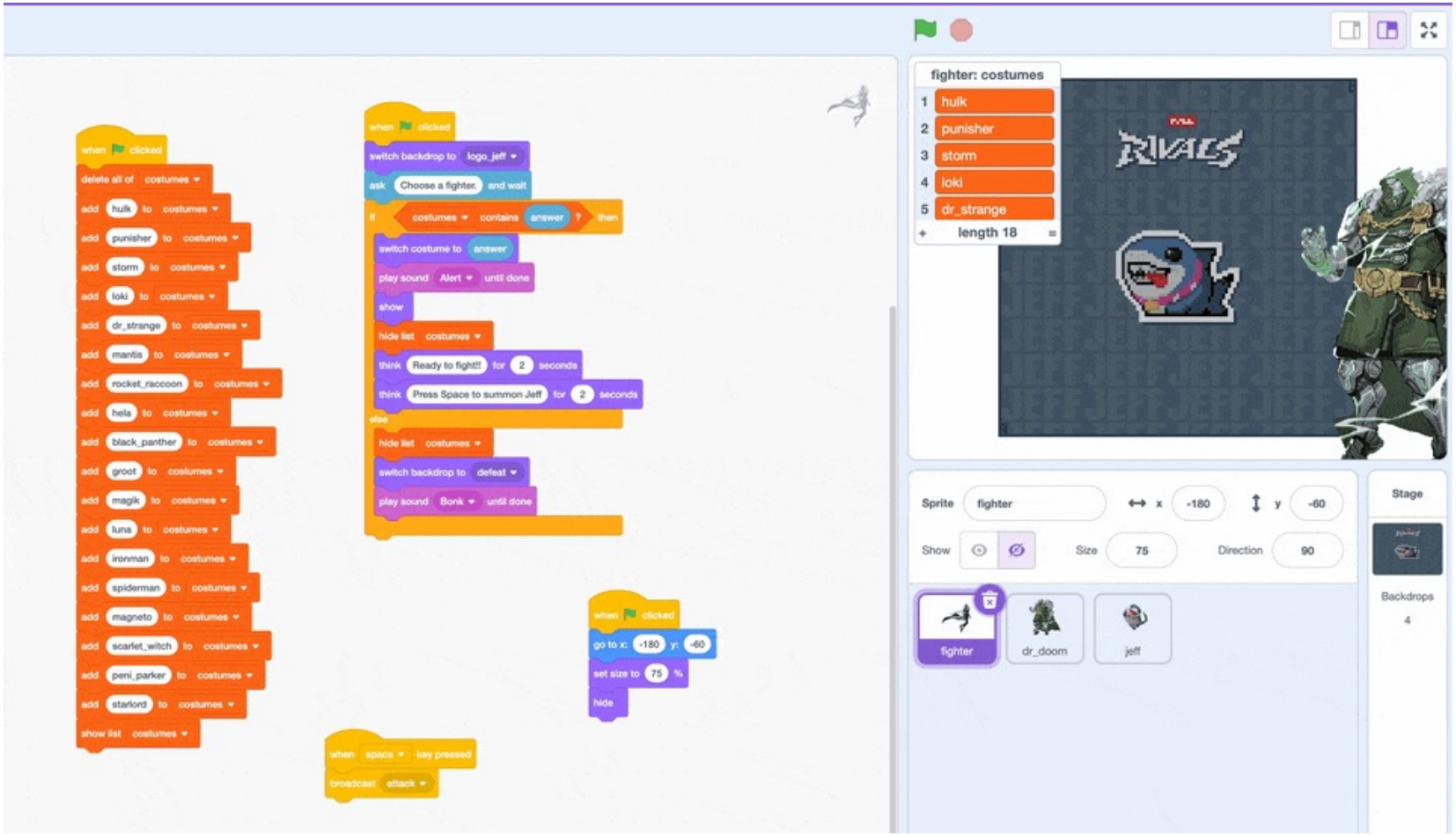
stage	code	**value** of temperature
e)		

Trace the **value** of temperature at each stage.

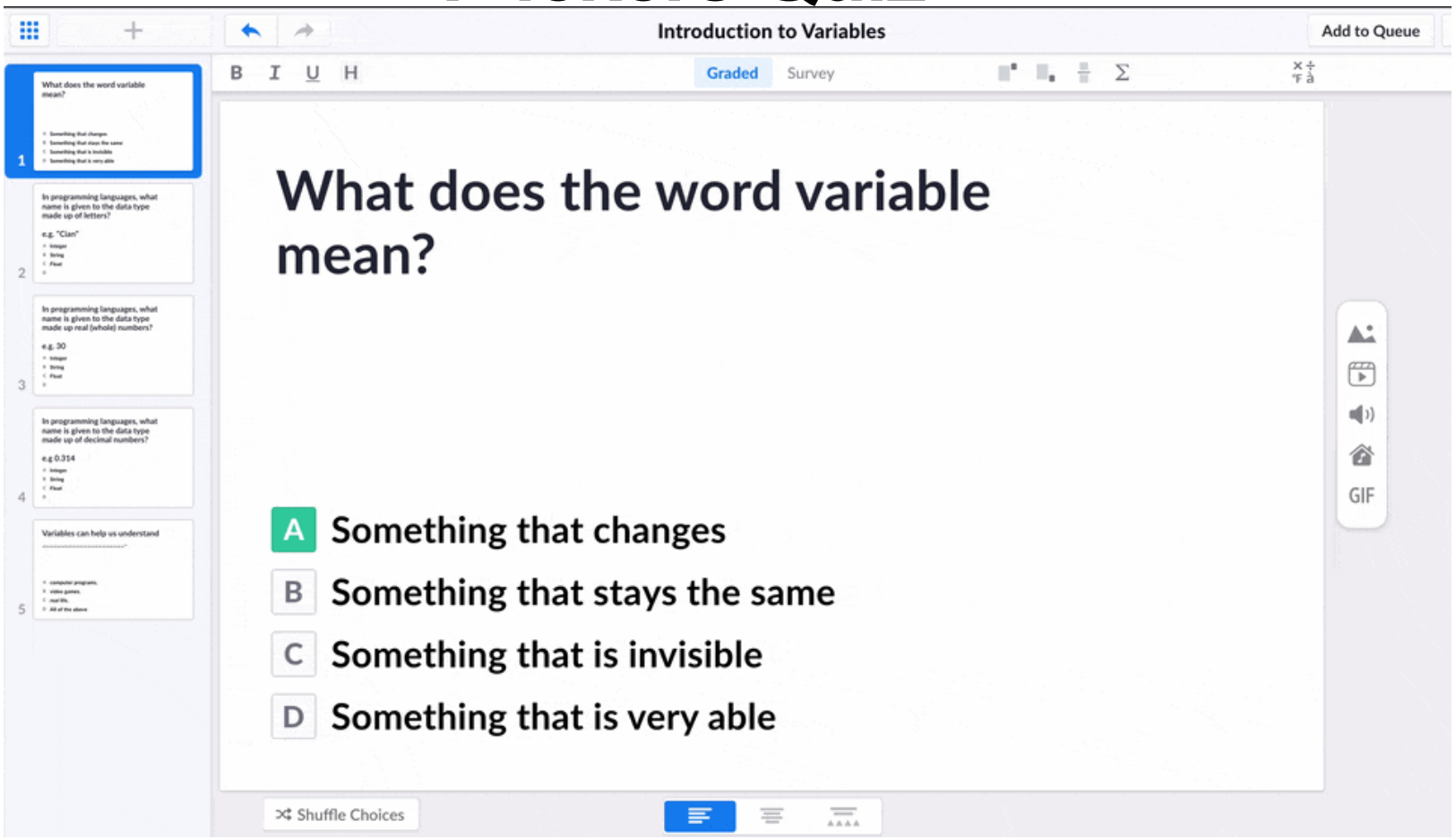


stage	**value** of temperature
a)	5
b)	10
c)	8
d)	0
e)	6

Worksheet Q4



Plickers Quiz



Recap

A **variable** is a named piece of data, stored in a memory location in a computer.

A **variable** can be **initialised** with a starting **value,** commonly of type integer, String or float.

If the **value** of a variable is updated, the previous value is **overwritten**.

A list can store multiple variables.

You can **trace** the value a variable holds during a program, this can help you find **bugs**.